

Results (Color Indices + Higher Order Color Indices)

Random Forest Classifier (with redshift)

Classification Report:

Test Accuracy: 0.9934957705658343

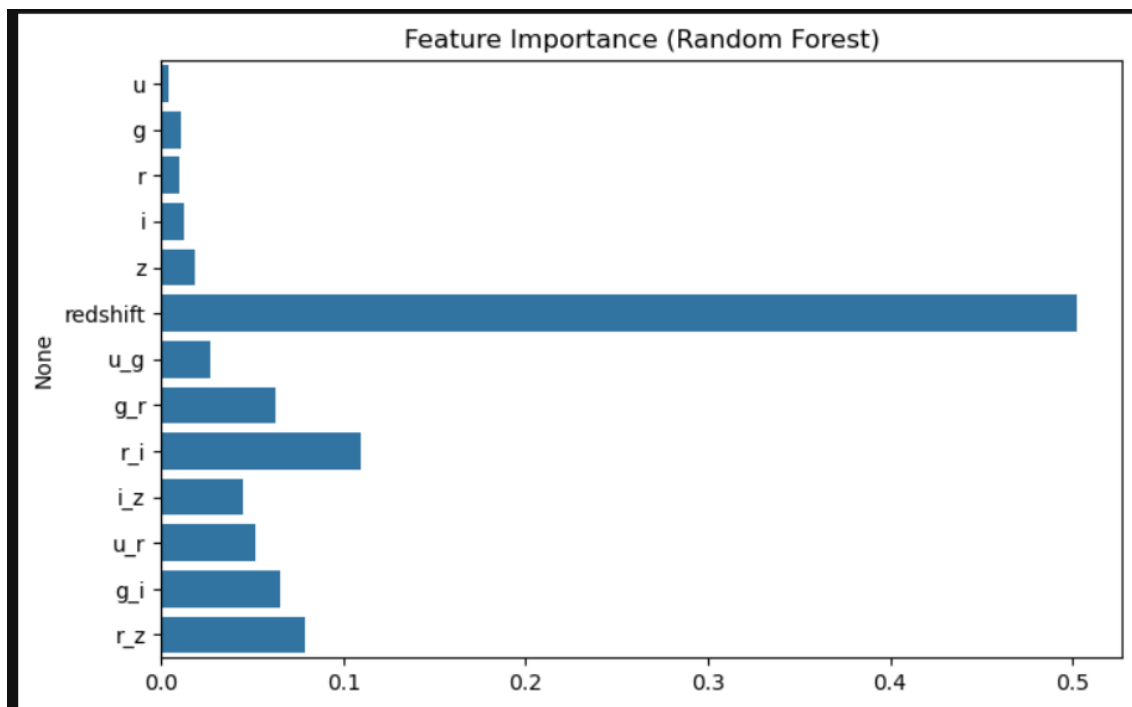
Confusion Matrix:

```
[[17522  39  17]
 [ 137 2629   0]
 [   0   0 9329]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	17578
1	0.99	0.95	0.97	2766
2	1.00	1.00	1.00	9329
accuracy			0.99	29673
macro avg	0.99	0.98	0.99	29673
weighted avg	0.99	0.99	0.99	29673

Feature Importances :

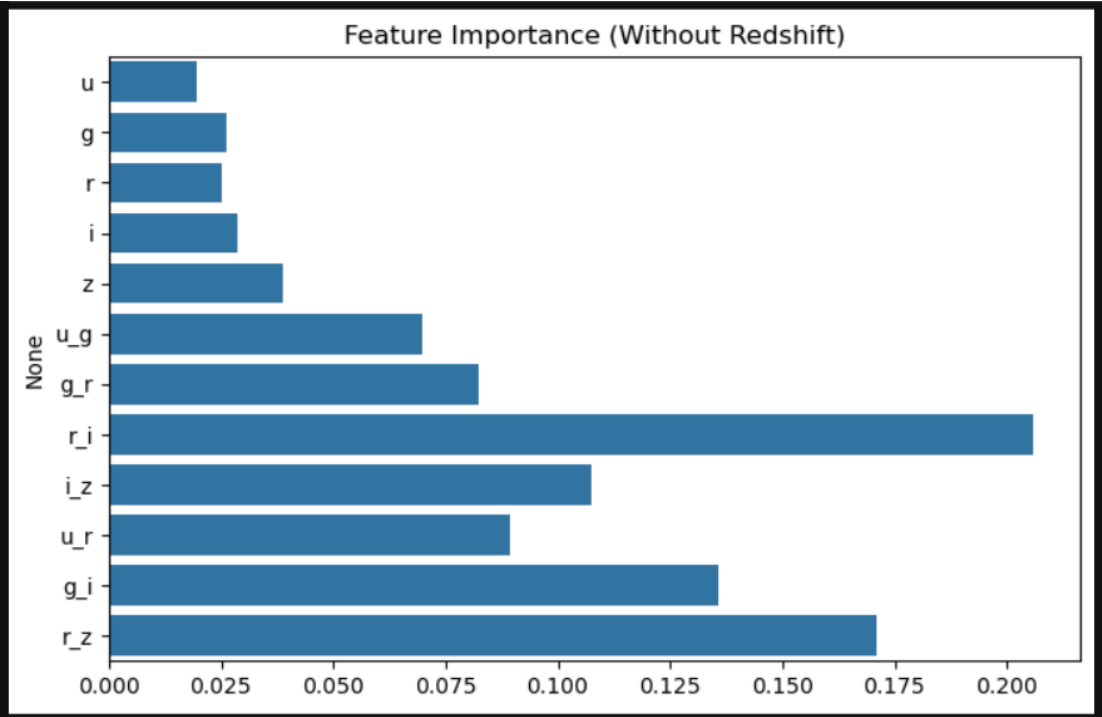


Random Forest (Without Redshift)

Classification Report:

Test Accuracy: 0.9287567822599669					
Confusion Matrix:					
[[17188 90 300]					
[161 2250 355]					
[933 275 8121]]					
Classification Report:					
	precision	recall	f1-score	support	
0	0.94	0.98	0.96	17578	
1	0.86	0.81	0.84	2766	
2	0.93	0.87	0.90	9329	
accuracy			0.93	29673	
macro avg	0.91	0.89	0.90	29673	
weighted avg	0.93	0.93	0.93	29673	

Feature Importances:



XGB Classifier (with redshift)

Classification Report :

Accuracy: 0.993765375930981

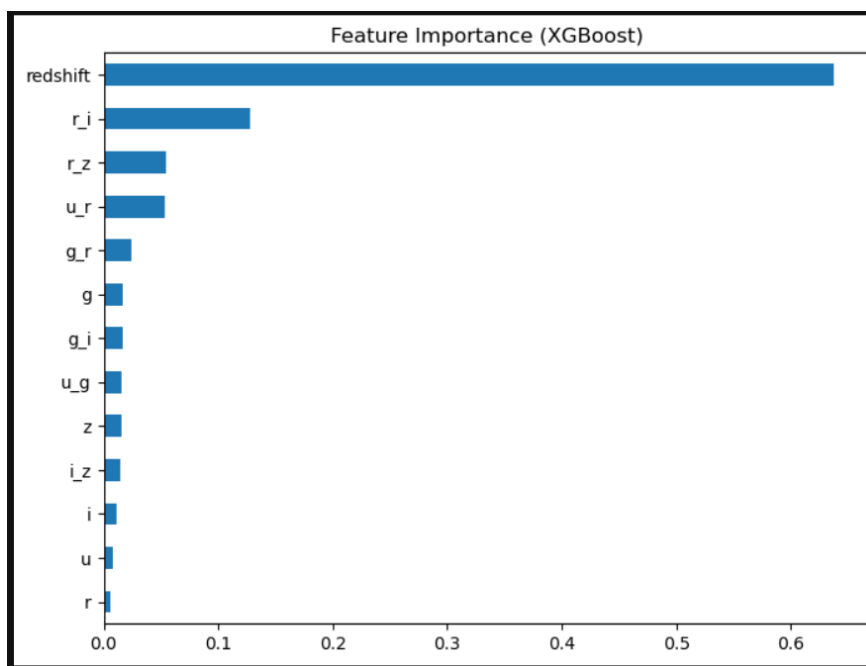
Confusion Matrix:

```
[[17518   35    25]
 [  122 2644     0]
 [     3     0 9326]]
```

Classification Report:

		precision	recall	f1-score	support
	0	0.99	1.00	0.99	17578
	1	0.99	0.96	0.97	2766
	2	1.00	1.00	1.00	9329
accuracy				0.99	29673
macro avg		0.99	0.98	0.99	29673
weighted avg		0.99	0.99	0.99	29673

Feature Importances:



XGB Classifier (without Redshift)

Classification Report:

Accuracy: 0.9337444815151822

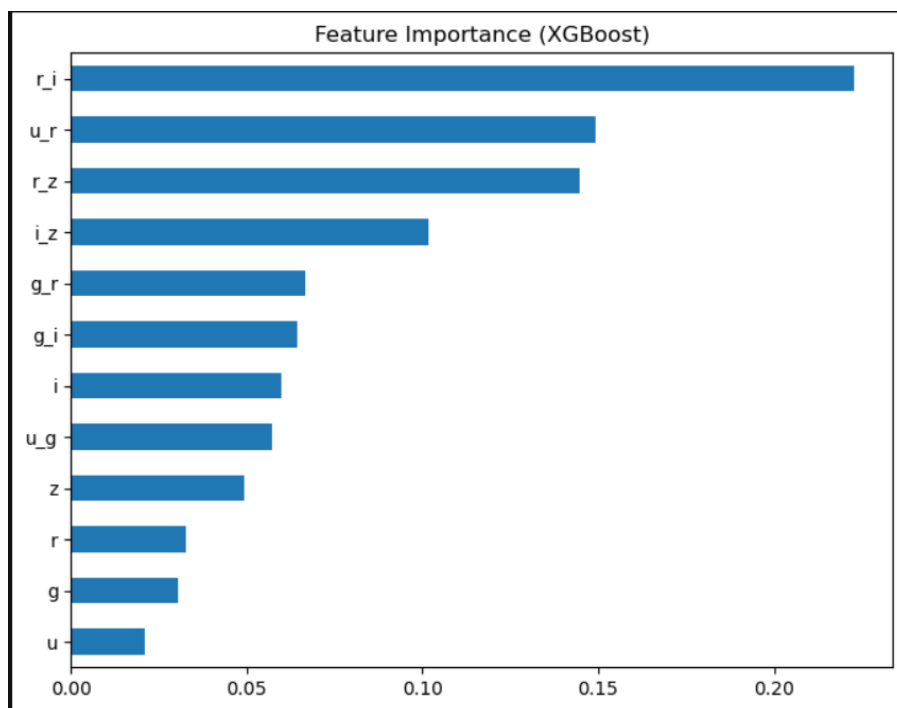
Confusion Matrix:

```
[[17137   91   350]
 [  147 2269   350]
 [   755   273 8301]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.95	0.97	0.96	17578
1	0.86	0.82	0.84	2766
2	0.92	0.89	0.91	9329
accuracy			0.93	29673
macro avg	0.91	0.90	0.90	29673
weighted avg	0.93	0.93	0.93	29673

Feature Importances:



FNN Keras (With Redshift)

Classification Report:

928/9281s 1ms/step

Confusion Matrix:

[[174676645]

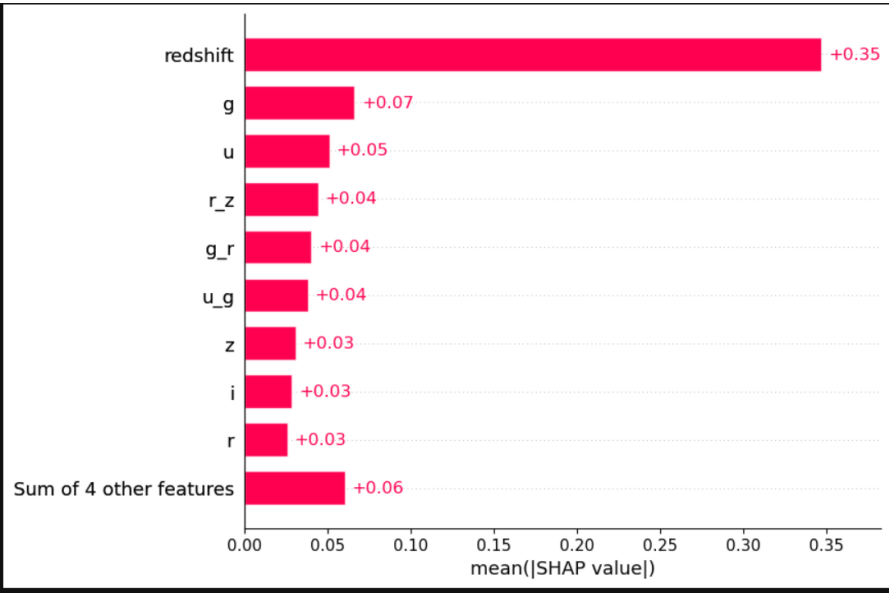
[11226540]

[10309226]]

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	17578
1	0.98	0.96	0.97	2766
2	1.00	0.99	0.99	9329
accuracy			0.99	29673
macro avg	0.99	0.98	0.98	29673
weighted avg	0.99	0.99	0.99	29673

Feature Importances:



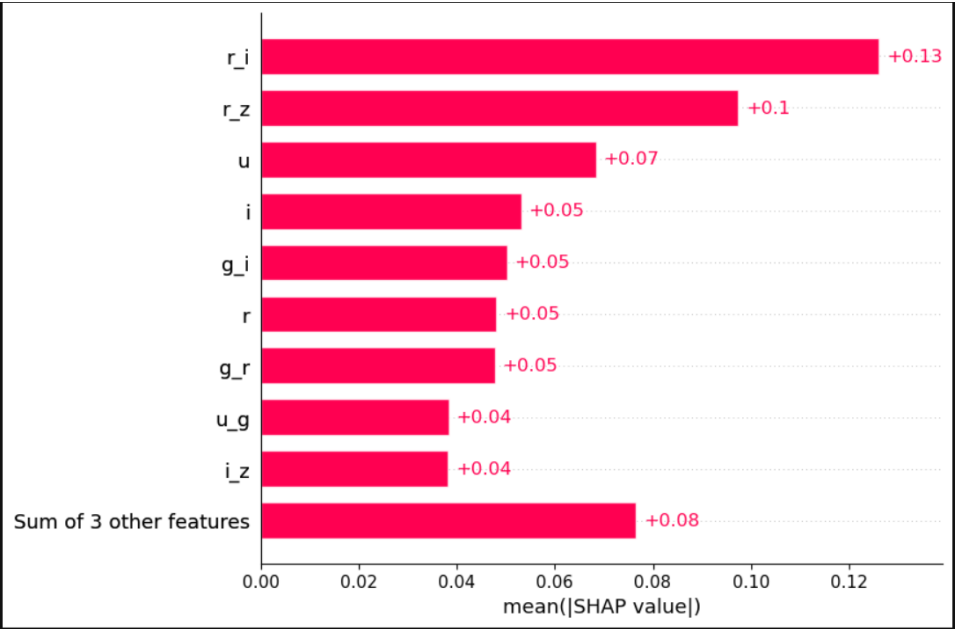
FNN Keras (Without Redshift)

Classification Report:

928/928 ————— 2s 2ms/step - accuracy: 0.9306 - loss: 0.2035
Test Accuracy: 0.9306440353393555

928/928 ————— 1s 1ms/step					
Confusion Matrix:					
[[17176 91 311]					
[159 2184 423]					
[857 217 8255]]					
Classification Report:					
	precision	recall	f1-score	support	
0	0.94	0.98	0.96	17578	
1	0.88	0.79	0.83	2766	
2	0.92	0.88	0.90	9329	
accuracy			0.93	29673	
macro avg	0.91	0.88	0.90	29673	
weighted avg	0.93	0.93	0.93	29673	

Feature Importances:



Stacking (FNN + XGB) (with redshift)

Classification Report:

K-Fold Stacking Accuracy: 0.9939675799548411

Confusion Matrix:

```
[[17511  46  21]
 [ 110 2656   0]
 [   2    0 9327]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	17578
1	0.98	0.96	0.97	2766
2	1.00	1.00	1.00	9329
accuracy			0.99	29673
macro avg	0.99	0.99	0.99	29673
weighted avg	0.99	0.99	0.99	29673

Agreement :

XGB Accuracy: 0.9939675799548411
FNN Accuracy: 0.990361608196003
Stacking Accuracy: 0.9939675799548411

Agreement %: 0.9939001786135544
Accuracy (agree): 0.9951851349518513
Accuracy (disagree): 0.7955801104972375

Stacking (FNN + XGB) (Without Redshift)

Classification Report:

```
K-Fold Stacking Accuracy: 0.9347555016344825
```

```
Confusion Matrix:
```

```
[[17079  104  395]
 [  133 2313  320]
 [   687  297 8345]]
```

```
Classification Report:
```

	precision	recall	f1-score	support
0	0.95	0.97	0.96	17578
1	0.85	0.84	0.84	2766
2	0.92	0.89	0.91	9329
accuracy			0.93	29673
macro avg	0.91	0.90	0.90	29673
weighted avg	0.93	0.93	0.93	29673

Agreement:

```
XGB Accuracy: 0.934317392916119
FNN Accuracy: 0.9258248239139959
Stacking Accuracy: 0.9347555016344825
```

```
Agreement %: 0.9579415630371044
Accuracy (agree): 0.950395778364116
Accuracy (disagree): 0.5785256410256411
```